

## GraphVideoAgent: LLM-Guided Dynamic Entity Graph Reasoning for Long Video Understanding

GraphVideoAgent (Meng et al., 2025) proposes a long-form video understanding system that builds a dynamic graph of entities and their relations across video frames, and uses an LLM agent to iteratively reason over this evolving graph to answer complex temporal and causal questions. It is specifically designed to handle sparse supervision and multimodal reasoning in long, egocentric or narrative-style videos.

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### Dynamic Entity-Relation Graph Construction (With Examples)

The system first uniformly samples a few frames and uses VLM tools (e.g., EVA-CLIP, LaViLa) to generate captions. NLP tools (e.g., spaCy) extract entities and their relations from the captions to build a graph:

- **Nodes:** unique entities (e.g., Dog, Person, Toy)
- **Edges:** labeled relations (e.g., plays\_with, takes, barks\_at), tagged with frame numbers and action types (spatial, interaction, or action)
- **States:** per-entity time-series (e.g., Dog: neutral → angry → barking)

### Example Sequence:

- Frame 1: “Dog plays with toy” → Dog --plays\_with--> Toy
- Frame 2: “Person takes toy from dog” → Person --takes--> Toy
- Frame 3: “Dog looks angry” → state update: Dog: angry
- Frame 4: “Dog barks at person” → Dog --barks\_at--> Person

The resulting graph tracks evolving relationships across time, enabling later reasoning about causality and temporal order.

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### LLM-Based Agent Reasoning (With Examples)

After building the initial graph, the system prompts an LLM (e.g., GPT-4) with the graph structure, question, and optionally captions. The LLM reasons over the graph to produce answers or request new frames.

### Example Input Prompt:

Question: Why did the dog bark at the person?  
Entities: Dog, Person, Toy  
Relations: Dog plays\_with Toy [F1], Person takes Toy [F2], Dog barks\_at Person [F4]  
States: Dog: neutral → angry [F3]  
Answer: null  
Confidence: 1 (insufficient)

### LLM Output:

```
{  
  "final_answer": "Because the person took the toy from the dog, which made it angry.",  
  "confidence": 3  
}
```

If the LLM is unsure, it self-reflects, suggests missing evidence (e.g., "Need more frames showing object interactions"), and triggers targeted frame retrieval, guided by graph-based relevance scoring.

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### Temporal Reasoning via Multi-Round Retrieval (With Examples)

If initial frames are insufficient, the LLM iteratively expands the graph by selecting more frames:

- Guided by graph-based scoring:  
$$\text{Score} = \alpha * \text{graph relevance} + \beta * \text{visual similarity} + \gamma * \text{temporal continuity}$$
- Each round updates the graph:  
$$G_{t+1} = U(G_t, \text{new\_frames}, \text{new\_relations})$$

This multi-round LLM-loop continues until the LLM reaches high confidence.

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### Evaluation and Results

GraphVideoAgent outperforms strong baselines on EgoSchema and NExT-QA benchmarks while using only ~8.2 frames per video (compared to 32–256 for others):

- EgoSchema: 56.3% (vs. 54.1% for VideoAgent, 55.6% for GPT-4V)

- NExT-QA: 73.3% (vs. 71.3% for VideoAgent, 67.7% for LLoVi)

Ablation studies confirm:

- Without entity relations → performance drops (−2.5%)
- Without temporal tracking → worse coherence
- Without graph → LLM alone struggles to model causality

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### Why It Matters for the Survey

GraphVideoAgent is one of the first systems to combine **LLMs with evolving scene graphs over time**, where the LLM acts not just as a reasoner but as an **adaptive control agent** that builds the graph, queries it, and decides what to retrieve next. The dynamic graph serves as memory, and the LLM provides the flexible inference engine. This makes it a key example of LLMs enabling structured reasoning over temporally-evolving graph-structured data.